

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.keaipublishing.com/jtte

Original Research Paper

On-ramp merging strategy for connected and automated vehicles based on complete information static game

Haigen Min ^{a,*}, Yukun Fang ^a, Xia Wu ^a, Guoyuan Wu ^b, Xiangmo Zhao ^{a,**}^a School of Information Engineering, Chang'an University, Xi'an 710064, China^b Department of Electrical and Computer Engineering, University of California, Riverside, CA 92521, USA

HIGHLIGHTS

- MS allocation is modeled as a two-player game, where Pontryagin's minimum principle is used for control law calculation.
- Illustrating in detail mixed-strategy Nash equilibrium of the game for MS allocation.
- Proposing a varying-scale grid search algorithm to search for candidate parameters in the cost function.

ARTICLE INFO

Article history:

Received 17 May 2021

Received in revised form

19 July 2021

Accepted 21 July 2021

Available online 29 July 2021

Keywords:

Connected and automated vehicles

On-ramp merging

Complete information static game

Optimal control

Varying-scale grid search

ABSTRACT

Improper handling of vehicle on-ramp merging may hinder traffic flow and contribute to lower fuel economy, while also increasing the risk of collisions. Cooperative control for connected and automated vehicles (CAVs) has the potential to significantly reduce negative environmental impact while also improve driving safety and traffic efficiency. Therefore, in this paper, we focus on the scenario of CAVs on-ramp merging and propose a centralized control method. Merging sequence (MS) allocation and motion planning are two key issues in this process. To deal with these problems, we first propose an MS allocation method based on a complete information static game whereby the mixed-strategy Nash equilibrium is calculated for an individual vehicle to select its strategy. The on-ramp merging problem is then formulated as a bi-objective (total fuel consumption and total travel time) optimization problem, to which optimal control based on Pontryagin's minimum principle (PMP) is applied to solve the motion planning issue. To determine the proper parameters in the bi-objective optimization problem, a varying-scale grid search method is proposed to explore possible solutions at different scales. In this method, an improved quicksort algorithm is designed to search for the Pareto front, and the (approximately) unbiased Pareto solution for the bi-objective optimization problem is finally determined as the optimal solution. The proposed on-ramp merging strategy is validated via numerical simulation,

* Corresponding author. Tel.: +86 29 6110 5011.

** Corresponding author. Tel.: +86 29 6110 5011.

E-mail addresses: hgmin@chd.edu.cn (H. Min), fangyukun@chd.edu.cn (Y. Fang), viichelle_wu@chd.edu.cn (X. Wu), gywu@cert.ucr.edu (G. Wu), xmzhao@chd.edu.cn (X. Zhao).

Peer review under responsibility of Periodical Offices of Chang'an University.

<https://doi.org/10.1016/j.jtte.2021.07.003>2095-7564/© 2021 Periodical Offices of Chang'an University. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

and comparison with other strategies demonstrates its effectiveness in terms of fuel economy and traffic efficiency.

© 2021 Periodical Offices of Chang'an University. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

An improper approach to handling vehicles on-ramp merging may lead to heavy traffic congestion, and in turn increase fuel consumption, travel time, and the risk of lateral collision (Chung and Recker, 2013; Iordanidou et al., 2014; Matarage and Dissanayake, 2020). On-ramps can lead to bottlenecks, since merging vehicles may slow down initially or even stop on the ramp as they wait for an opportunity to merge. Collaborative control allows connected and automated vehicles (CAVs) to cooperate with each other through advanced localization (Min et al., 2019; Zhao et al., 2019), perception (Zhao et al., 2020), and communication technologies (such as V2V (vehicle to vehicle) and V2I (vehicle to infrastructure) communication (Kabashkin, 2017; Yang et al., 2018)) to provide them with the requisite information for decision making. And, advanced algorithms that control the CAVs on-ramp merging have the potential to improve traffic efficiency, fuel economy, and safety (Altan et al., 2017; Li et al., 2017; Massera Filho et al., 2017; Moradi-Pari et al., 2017).

The allocation of the merging sequence (MS) and vehicle motion planning are the two main issues in handling on-ramp merging for CAVs (Jing et al., 2019). MS reflects the priority a vehicle is given to leave the ramp, and it directly influences the motion planning process. For the motion planning algorithm, it should first ensure the safety of all vehicles and then control the vehicles passing the on-ramp according to the MS. These two aspects jointly influence the traffic efficiency and fuel economy, and in this paper, we propose a MS allocation approach based on game theory and adopt optimal control for motion planning. Specifically, a complete information static game is utilized to calculate MS, and the motion planning issue is formulated as a bi-objective optimization problem.

In a complete information static game, players simultaneously select their strategies (Orbay and Sevgi, 2014). In this case, “simultaneously” means that players do not know how others will maneuver when they make choices from their strategy set. Since we assume all players are rational in game theory (Myerson, 1997), mixed-strategy Nash equilibrium forms the basis for a player to choose its approach, and investigating this is of great significance for the analysis of the behaviors of individuals in a game. However, in the existing literature related to on-ramp merging using game theory, we rarely see a detailed illustration of this equilibrium. In addition, when determining the final solution from the Pareto front in a bi-objective or multi-objective optimization problem, the

standard can be very subjective. Considering these two issues, the main contributions of this study are as following.

- 1) Illustrating in detail mixed-strategy Nash equilibrium of the complete information static game for MS allocation, with the equilibrium utilized for an individual vehicle to select its strategy.
- 2) Proposing a varying-scale grid search algorithm to numerically search for acceptable parameters in the cost function of the bi-objective optimization problem, where the scale for searching is progressively narrowed down until the solution's precision is satisfied.
- 3) Adopting a modified quicksort algorithm to find the Pareto front, and then the (approximately) unbiased Pareto solution (Wang et al., 2017) is mathematically derived. Theoretically, the solution we obtain is optimal if the two objectives in the bi-objective optimization problem are considered to be equally important.

To simplify our analysis, we model the MS allocation process as a two-player game. Pontryagin's minimum principle is applied to calculate the optimal control law for the bi-objective optimization problem, and the coefficient parameters of different objectives in the objective function are determined by searching the unbiased Pareto solution.

The rest of this paper is structured as follows. Related work is introduced in section 2, and section 3 formulates the problem and describes the methodology. Simulation results are presented in section 4, and we briefly summarize our work in section 5.

2. Related work

The majority of the current researches about on-ramp merging emphasizes MS allocation (Chen and Englund, 2015; Morales and Nijmeijer, 2016) and motion planning algorithms that allow the vehicles to smoothly pass the merging point while improving fuel economy and traffic efficiency (Cao et al., 2015; Lin et al., 2020; Ntousakis et al., 2014, 2016; Rios-Torres and Malikopoulos, 2016; Triest et al., 2020). MS allocation is basically a scheduling problem that determines the passing order for each vehicle. Chen et al. (2020) divided the approaches to establishing merging sequence into “rule-based methods” and “optimal methods”. Rule-based methods depend on explicit rules and typically include virtual mapping (Wang et al., 2013) and first-in-first-out (FIFO) approaches (Rios-Torres and Malikopoulos, 2017), among others. Optimal

methods are based on a global or local performance evaluator to assess the expected benefits brought by a possible MS. Currently, reinforcement learning based researches about on-ramp merging are gradually popularized. For example, [Triest et al. \(2020\)](#) applied a hierarchical method for decision making, where a high-level policy for on-ramp merging were trained using reinforcement learning and the output of the policy was executed via the low-level controller.

Among the optimal methods, those based on game theory are increasingly found in the literature. Game theory is the study of mathematical models of conflict and cooperation between decision-makers, and it is therefore appropriate to analyze the competition and cooperation between CAVs to determine MS. In fact, CAVs merging on on-ramps can be regarded as a competition for the priority to pass the ramp from the perspective of an individual agent, with its priority reflected in the merging sequence. Solutions to this require the cooperation of each agent, and the result of MS allocation is also a reflection of compromise between agents. [Kang and Rakha \(2017\)](#) proposed a decision-making process for merging maneuvers based on non-cooperative game theory, whereby the leading vehicle chooses to keep its current state, and the lag vehicle (on a different road) either remains on its original state or surpasses the leading vehicle. [Jing et al. \(2019\)](#) proposed an optimization framework based on a cooperative multi-player game to coordinate vehicles and achieve maximum global pay-off, including fuel economy, traffic efficiency, and driving comfort. The authors decomposed the multi-player game into multiple two-player games to allocate an MS and formulated the on-ramp merging problem as an optimization problem to deal with the issue of motion planning.

Once an MS is ascertained, motion planning algorithms are required to ensure that vehicles smoothly pass the ramp without colliding, while also achieving enhanced traffic performance goals such as fuel economy or traffic efficiency. Generally speaking, the control algorithms governing on-ramp CAVs merging can be divided into centralized methods ([Awal et al., 2013](#); [Raravi et al., 2007](#)) and decentralized methods ([Cao et al., 2015](#); [Wang et al., 2018](#)). In centralized methods, a single central controller globally plans the trajectory of all vehicles, while in decentralized methods, each vehicle makes its own decision in accordance with the information it receives from other vehicles on the road, or other sources of information ([Rios-Torres and Malikopoulos, 2017](#)) such as a roadside unit. [Cao et al. \(2015\)](#) presented a centralized cooperative merging method, in which vehicles on the main road can obtain information about vehicles on the ramp road via the centralized controller, so that they can adjust their state to merge cooperatively at the ramp and optimize passing efficiency. [Ntousakis et al. \(2016\)](#) proposed a decentralized automatic merging algorithm whereby each vehicle utilizes the information it receives from other vehicles via V2I and V2V communication to handle the available gaps between vehicles and smoothly pass the ramp. Experiments showed that this algorithm could cooperatively coordinate the vehicles so they passed the ramp without collisions, while also maintaining traffic stability. [Wang et al. \(2018\)](#) developed a novel distributed consensus protocol for CAVs to cooperate with

each other in the on-ramp merging scenario using V2V and V2I communication, where vehicles applied this protocol to adapt their speeds and positions to the preceding vehicles (either physical ones on the same lane or “ghost” ones projected from the other lane) so that all vehicles could cooperatively pass the merging area. Further, some researchers have adopted both centralized and decentralized methods to deal with the motion planning problem. For example, a trajectory planning approach ([Ntousakis et al., 2016](#)) combining both centralized and decentralized methods were proposed to optimize engine efficiency and passenger comfort by adding jerk (derivative of acceleration) in the objective function, where the analytical solution was obtained using optimal control theory combined with the linear quadratic regulator method, and a model predictive control scheme was used to compensate for potential interference. Currently, learning based method especially reinforcement learning (RL) is emerging in the literature for motion planning of on-ramp merging research. [Lin et al. \(2020\)](#) employed deep reinforcement learning for decentralized decision-making and longitudinal control for high-speed on-ramp merging, where deep deterministic policy gradient (DDPG) is the algorithm for training to output continuous control actions. Simulation showed that, by adding a small jerk penalty in the multi-objective reward function, the vehicle jerk (related to the passenger comfort) could be reduced by 73% and the collision rate was maintained at zero in the on-ramp merging scenario. [Hu et al. \(2019\)](#) proposed an actor-critic-based approach, where they incorporated the low-level controllers into the reinforcement learning algorithms via a masking mechanism. At a given state, feasible actions were determined via a low-level controller.

3. Problem statement and methodology

3.1. Scenario description

We consider on-ramp merging in a scenario that consists of a single lane of a main road and a ramp road, as shown in [Fig. 1](#). Usually, the main road would consist of multiple lanes; however, in this paper we focus on the merging scenario and only take into consideration the lane that is in conjunction with the on-ramp. We assume that there is a centralized controller that can communicate with all vehicles in the control area without any time delay, and each vehicle can obtain the control signal sent by the controller. The region described in [Fig. 1](#) is divided into control area (with length L), where vehicles adjust their speeds under the control of the centralized controller to prepare for the merge, and merging area (with length M), where potential lateral collision of the vehicles may occur. Due to the control effect of the centralized controller, all vehicles can travel through the merging area at the expected velocity v_m and the safe passage of vehicles at the merging point (one at a time) can also be guaranteed. A complete information static game is conducted when a vehicle reaches the beginning of the control area, which is noted as

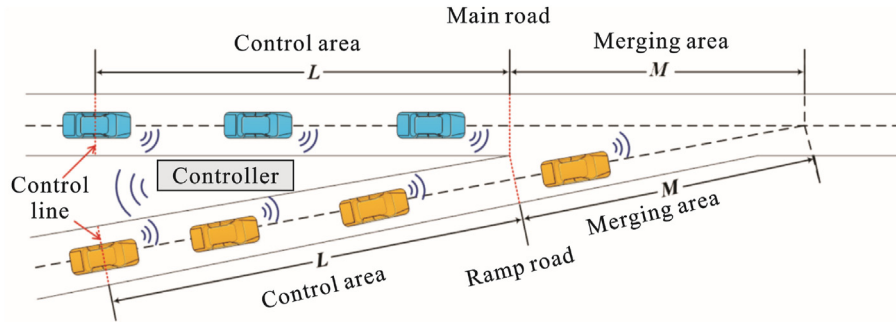


Fig. 1 – General description of the on-ramp merging scenario.

the control line. Details of the game are illustrated in the next part of this section.

There are some assumptions as follows:

- Overtaking is not allowed for vehicles on either the main road or the ramp road since we focus on the merging scenario in this paper especially the performance of merging sequence method and vehicle motion planning algorithm. Thus, we just take into consideration the lane (on the main road) that is in conjunction with the ramp road to refine the process of modeling.
- Lateral control is not considered because we constrain that only one vehicle is in the merging area at a time.
- The centralized controller can communicate with all vehicles in the control area without any time delay and each vehicle can obtain the information shared by the controller instantly. Though communication delay does exist in practice, we focus on MS allocation and motion planning and simply ignore the impact caused by communication delay to simplify our model.
- All vehicles in our scenario are homogeneous and each vehicle is modeled as a mass point. Because of the homogeneity assumption, difference of the internal delay, like actuator lag, among vehicles can be ignored (Wang, 2018), which simplifies the modeling.

For each vehicle V_i , $i = 1, 2, \dots$, its state is depicted as

$$\chi_i(t) = [p_i(t), v_i(t), a_i(t)] \quad (1)$$

where $p_i(t)$, $v_i(t)$, $a_i(t)$ represent the position, velocity, and acceleration of vehicle V_i at moment t , respectively. The moment at which vehicle V_i arrives at the control line is noted as t_i^0 , and the moment when it leaves the control area is noted as t_i^f . The state of the vehicle V_i at the control line is noted as $\chi_i^0 = \chi_i(t_i^0)$ with components $p_i^0 = p_i(t_i^0)$, $v_i^0 = v_i(t_i^0)$, $a_i^0 = a_i(t_i^0)$.

3.2. Determination of merging sequence based on complete information static game

MS determination is the precondition for global motion planning, and a complete information static game is applied to ascertain the MS for each vehicle in this paper. To simplify the game's structure, it is modeled as a two-player complete information static game, as depicted in Fig. 2.

We consider an increasing number of CAVs entering the control area (Fig. 1). Since overtaking is not allowed, the game is performed among vehicles on two different roads. The game begins when a vehicle arrives at the control line. For example, in Fig. 2(a), vehicle V_m reaches the control line and a game between V_m and V_i occurs. If the result of the game

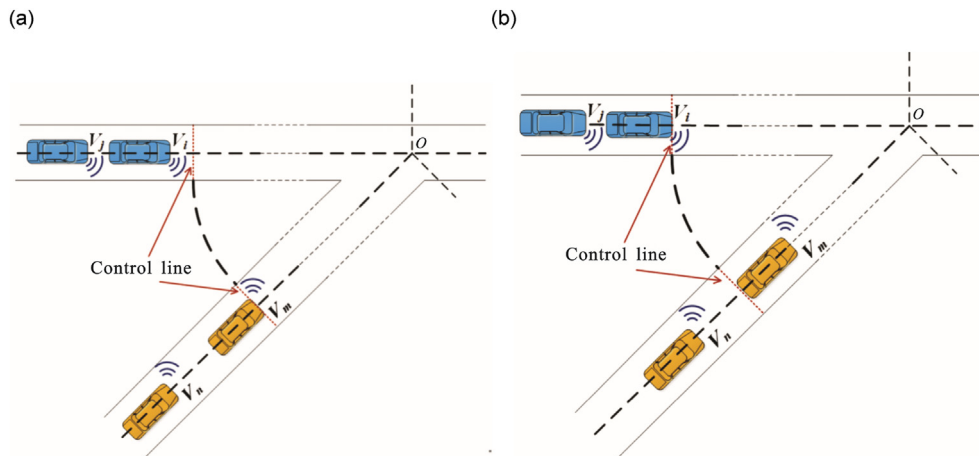


Fig. 2 – Illustration of the game scenario. (a) A game between V_m and V_i . (b) A game between V_n and V_i .

is that V_m becomes the preceding vehicle, V_m gets the minimum unallocated MS. If the result of the game is that V_m becomes the following vehicle, V_i gets the minimum unallocated MS and another game between V_m and V_j is activated. This procedure will continue until V_m obtains an MS. Assuming V_m becomes the preceding vehicle after the game between V_m and V_i , a new game between V_i and V_n occurs when V_i reaches the control line, shown in Fig. 2(b). For each vehicle, an MS is obtained similarly. All possible passing sequences for the scenario shown in Fig. 2 are $V_m-V_n-V_i-V_j$, $V_m-V_i-V_n-V_j$, $V_m-V_i-V_j-V_n$, $V_i-V_j-V_m-V_n$, $V_i-V_m-V_j-V_n$, $V_i-V_m-V_n-V_j$.

For a two-player game $G = \{S_i, u_i | i = 1, 2\}$, where S_i represents the strategy set of player i , $s_i \in S_i$ is a specific strategy applied by player i . In the context of this paper, there are only two specific strategies: a vehicle chooses to be the preceding vehicle (leader) or the following vehicle (follower). u_i represents the pay-off of player i and $u_i(s_i)$ represents the pay-off of player i selecting the specific strategy s_i . The pay-off matrix for this two-player game is shown in Table 1. Player 1 is the vehicle reaching the control line and player 2 is the counterpart.

In Table 1, the first value in the brackets represents the pay-off of player 1 and the second value in the brackets represents the pay-off of player 2. θ and λ are two probability values. Player 1 chooses to be the leader with probability θ , and to be the follower with probability $1-\theta$. Player 2 chooses to be the leader with probability λ , and to be the follower with probability $1-\lambda$. c represents the collision risk, which is related to the vehicle's state at the moment the game occurs. If player 1 and player 2 simultaneously choose to be the leader or the follower, which would mean they were competing for the same MS, they would collide with each other at the merging point. Note the moment when a game occurs as t_g , and the states for player 1 and player 2 are $\chi_1(t_g)$ and $\chi_2(t_g)$, respectively. The collision risk is defined as

$$c = \frac{1}{\|\chi_1(t_g) - \chi_2(t_g)\|} \quad (2)$$

where $\|\cdot\|$ is 2-norm. This equation indicates that the collision risk is high if the state of the two players is similar. m_1 , m_2 , n_1 , n_2 are the pay-off of each player and are related to their fuel consumption and travel time. According to Jing et al. (2019), we define the cost function (J_i) for each vehicle V_i as

$$J_i = \frac{1}{2} \int_{t_0^i}^{t_f^i} w_1 a_i^2(t) dt + w_2 T_i \quad (3)$$

where $a_i(t)$ represents the acceleration of vehicle V_i at moment t , T_i is the travel time (from moment t_0^i to moment t_f^i). For the

cost function, the first item is surrogate of fuel consumption and the second item is related to travel time. w_1 and w_2 are coefficient parameters that measure the importance of fuel consumption and travel time respectively. Since Eq. (3) contains factors of time, distance, and acceleration, and the units and value scale of these are different, we normalize each item to the range $[0, 1]$ to avoid any factor that dominates the others. The same principle is also applied in the subsequent text when it comes to summations of dual/multiple objectives.

Since the determination of $a_i(t)$ and T_i is related to content about optimal control in subsequent parts of this section, we will illustrate later how to define m_1 , m_2 , n_1 , n_2 and the way to select w_1 and w_2 . The rest of this part will present the derivation process of obtaining the mixed-strategy Nash equilibrium for the game G .

For a complete information static game, the mixed-strategy Nash equilibrium always exists (Myerson, 1997). According to the pay-off matrix, the expected pay-off for player 1 is

$$u_1 = -c\theta\lambda + m_1\theta(1-\lambda) + n_1(1-\theta)\lambda - c(1-\theta)(1-\lambda) \quad (4)$$

For player 2, the expected pay-off is

$$u_2 = -c\theta\lambda + m_2\theta(1-\lambda) + n_2(1-\theta)\lambda - c(1-\theta)(1-\lambda) \quad (5)$$

Player 1 is expected to adopt the best probability distribution θ^* to maximize its pay-off u_1 . Similarly, player 2 is expected to adopt the best probability distribution λ^* to maximize its expected pay-off u_2 . Differentiating the pay-off with the probability distribution, respectively, we can get

$$\frac{\partial u_1}{\partial \theta} = (-2c - m_1 - n_1)\lambda + m_1 + c \quad (6)$$

$$\frac{\partial u_2}{\partial \lambda} = (-2c - m_2 - n_2)\theta + n_2 + c \quad (7)$$

The stationary points for Eqs. (6) and (7) are

$$\begin{cases} \theta^* = \frac{n_2 + c}{2c + m_2 + n_2} \\ \lambda^* = \frac{m_1 + c}{2c + m_1 + n_1} \end{cases} \quad (8)$$

where θ^* and λ^* are the best probability distribution for player 1 and player 2 to select their strategies, which means that the only mixed-strategy Nash equilibrium for this game G is $[(\theta^*, 1-\theta^*), (\lambda^*, 1-\lambda^*)]$.

The principle for such a game can be concluded as: "each player selects a strategy that can lead to maximum pay-off, considering the worst situation". However, the equilibrium is not a pure strategy but a probability distribution, and it is likely that both players choose to be the leader or the follower. The reason for such a conflict is usually because the states of two players are very similar. According to Eq. (2), if $\chi_1(t_g) \rightarrow \chi_2(t_g)$, $c \rightarrow \infty$, and then $\theta^* \rightarrow 1/2$, $\lambda^* \rightarrow 1/2$, which means both players randomly choose their strategy in such a situation. To avoid ambiguity, if both players choose to be the leader or follower, player 1 becomes the leader and player 2 becomes the follower, since the position of player 1 is closer to merging point O.

Table 1 – Pay-off matrix for the two-player based on complete information static game.

Player 1	Player 2	
	Leader(λ)	Follower($1-\lambda$)
Leader(θ)	$(-c, -c)$	(m_1, m_2)
Follower($1-\theta$)	(n_1, n_2)	$(-c, -c)$

3.3. Optimal control for motion planning

Motion planning is another crucial aspect of the on-ramp merging problem. Eq. (3) has given the cost function for a single vehicle; here we formulate the motion planning problem as a global optimization problem.

$$\min_{a_i} J_i = \min_{a_i} \left[\frac{1}{2} \sum_{i=1}^W \int_{t_i^0}^{t_i^f} w_1 a_i^2(t) dt + \sum_{i=1}^W w_2 T_i \right] \quad (9)$$

Subject to

$$a_{\min} \leq a_i(t) \leq a_{\max}, 0 \leq v_i(t) \leq v_{\max}, \forall t \in [t_i^0, t_i^f] \quad (10)$$

where $a_{\min}$ and $a_{\max}$ are the limitations for acceleration, $v_{\max}$ is the limitation for velocity, and W is the number of vehicles during the research time. The travel time of each vehicle can be presented as

$$T_i = t_i^f - t_i^0 \quad (11)$$

Note that the subscript i represents the passing order (at the merging point) of vehicle V_i . The final control effect for the motion planning of each vehicle is that each vehicle traverses the merging area at the same velocity v_m while keeping a safe distance δ , thus, the relationship between t_i^f and t_{i-1}^f can be formulated as

$$t_i^f = t_{i-1}^f + \frac{\delta}{v_m} \quad (12)$$

Furthermore, we put a constraint that only one vehicle can be in the merging area at a time to avoid potential lateral collision, which means that vehicle V_i can enter the merging area only after vehicle V_{i-1} exits. Therefore, the distance between V_{i-1} and V_i is the length of the merging area M , and Eq. (12) can be rewritten as

$$t_i^f = t_{i-1}^f + \frac{M}{v_m} \quad (13)$$

Thus, if t_i^f is determined, all t_i^f would be determined.

We need to determine the acceleration of each vehicle a_i as the control law for motion planning, and in this paper, we adopt the optimal control methods based on Pontryagin's minimum principle to calculate the optimal control law (Rios-Torres and Malikopoulos, 2017). From Eq. (9) and the relationship between position, velocity, and acceleration (velocity is the derivative of position, and acceleration is the derivative of velocity), the Hamiltonian function for each vehicle V_i can be formulated as

$$H_i(t, \chi_i(t), a_i(t)) = \frac{1}{2} a_i^2 + \lambda_i^p v_i + \lambda_i^v a_i \quad (14)$$

where λ_i^p and λ_i^v are the co-state components. To guarantee the optimality, let

$$\frac{\partial H_i}{\partial a_i} = a_i + \lambda_i^v = 0 \quad (15)$$

satisfying the Euler-lagrange equation

$$\dot{\lambda}_i^p = -\frac{\partial H_i}{\partial p_i} = 0 \quad (16)$$

$$\dot{\lambda}_i^v = -\frac{\partial H_i}{\partial v_i} = -\lambda_i^p \quad (17)$$

Solving Eqs. (16) and (17), we have $\lambda_i^p = b_i$ and $\lambda_i^v = -(b_i t + c_i)$, where b_i, c_i are the constants when solving the equations of the corresponding vehicle V_i . Combining the optimality condition given in Eq. (15), the optimal control law is

$$a_i^*(t) = b_i t + c_i \quad (18)$$

Further, we can get the velocity function $v_i^*(t)$ and position function $p_i^*(t)$ under the optimal control law.

$$v_i^*(t) = \frac{1}{2} b_i t^2 + c_i t + d_i \quad (19)$$

$$p_i^*(t) = \frac{1}{6} b_i t^3 + \frac{1}{2} c_i t^2 + d_i t + e_i \quad (20)$$

where b_i, c_i, d_i, e_i are the coefficients when calculating the integration. If the initial value conditions and final value conditions are given, coefficients in Eqs. (18)–(20) can be obtained and then the optimal control law can be determined. Combining the initial value conditions and final value conditions, these coefficients can be obtained by solving the matrix Eq. (21).

$$\begin{bmatrix} \frac{1}{6}(t_i^0)^3 & \frac{1}{2}(t_i^0)^2 & t_i^0 & 1 \\ \frac{1}{2}(t_i^0)^2 & t_i^0 & 1 & 0 \\ \frac{1}{6}(t_i^f)^3 & \frac{1}{2}(t_i^f)^2 & t_i^f & 1 \\ \frac{1}{2}(t_i^f)^2 & t_i^f & 1 & 0 \end{bmatrix} \begin{bmatrix} b_i \\ c_i \\ d_i \\ e_i \end{bmatrix} = \begin{bmatrix} p_i(t_i^0) \\ v_i(t_i^0) \\ p_i(t_i^f) \\ v_i(t_i^f) \end{bmatrix} \quad (21)$$

where t_i^f can be calculated through Eq. (13), $p_i(t_i^0)$ is the position (or coordinate value) of the control line and $p_i(t_i^f)$ is the terminal position (or coordinate value) of the control area, $v_i(t_i^0)$ is the known velocity of the vehicle V_i at the moment t_i^0 , $v_i(t_i^f)$ is the expected velocity to pass the merging area and $v_i(t_i^f)$ is equal to v_m . Once these initial and final value conditions are given, coefficients b_i, c_i, d_i, e_i can be confirmed for the corresponding vehicle V_i .

3.4. Definition of the pay-off for each player

Here, we will define the pay-off for each player in the pay-off matrix. In other words, we will define the way of calculating m_1, m_2, n_1, n_2 shown in Table 1.

As stated above, m_1, m_2, n_1, n_2 are related to the cost function for each vehicle (Eq. (3)). Note the subscript i as the allocated sequence for vehicles V_i . For different MS, the values of $a_i(t), t_i^0, t_i^f$ differ from others. Therefore, the way to determine the MS will influence the cost function. In this paper, we use the FIFO model as the benchmark. For FIFO model, an MS is allocated only in accordance with the moment that each vehicle reaches the control line; i.e., the earlier the vehicle reaches the control line, the higher the priority that its MS order will be allocated. The MS for vehicle V_i determined by the FIFO model is noted as FIFO(i),

while by the proposed model, it is noted as GAME(i). GAME(i) has two types: the vehicle that is a leader is GAME_{leader}(i), while the follower is GAME_{follower}(i). The subscripts for player 1 and player 2 are i_1 and i_2 , respectively, and m_1 is defined as

$$m_1 = w_1 \frac{1 / \int_{t_0^{\text{GAME}_{\text{leader}}(i_1)}}^{t_1^{\text{GAME}_{\text{leader}}(i_1)}} a_{\text{GAME}_{\text{leader}}(i_1)}^2(t) dt}{1 / \int_{t_0^{\text{FIFO}(i_1)}}^{t_1^{\text{FIFO}(i_1)}} a_{\text{FIFO}(i_1)}^2(t) dt} + w_2 \frac{1 / T_{\text{GAME}_{\text{leader}}(i_1)}}{1 / T_{\text{FIFO}(i_1)}} \quad (22)$$

The first item is related to the benefit of fuel consumption and the second item depicts the benefit of travel time. Such definition considers the comparison with the benchmark (i.e., the FIFO model in this paper) from fuel consumption and travel time in the perspective of a single vehicle. For a vehicle, less fuel consumption means higher benefits, which means there is an inverse relation between the pay-off and the calculated surrogate fuel consumption (integral terms in the first item). Similarly, less travel time means higher benefits and the inverse relationship between the pay-off and the travel time also holds. Because of the inverse relationships, m_1 is defined as the formation like Eq. (22), and it can be simplified as

$$m_1 = w_1 \frac{\int_{t_0^{\text{FIFO}(i_1)}}^{t_1^{\text{FIFO}(i_1)}} a_{\text{FIFO}(i_1)}^2(t) dt}{\int_{t_0^{\text{GAME}_{\text{leader}}(i_1)}}^{t_1^{\text{GAME}_{\text{leader}}(i_1)}} a_{\text{GAME}_{\text{leader}}(i_1)}^2(t) dt} + w_2 \frac{T_{\text{FIFO}(i_1)}}{T_{\text{GAME}_{\text{leader}}(i_1)}} \quad (23)$$

Taking the first item in Eq. (23) as an example, if the denominator is less than the numerator, it means player 1 consumes less fuel in the proposed model than that in the FIFO model as being the leader. Therefore, the value of this item would be greater than one, which means player 1 gains more benefit from fuel consumption in this case. On the contrary, if the denominator is greater than the numerator, it means player 1 (potentially) consumes more fuel in the proposed model than that in the FIFO model as being the leader. The value of this item would be less than one, which means player 1 gains less benefit from fuel consumption in such case. Analysis for the second items in Eq. (23) is the same.

Similarly, m_2 , n_1 and n_2 can be defined as

$$m_2 = w_1 \frac{\int_{t_0^{\text{FIFO}(i_2)}}^{t_1^{\text{FIFO}(i_2)}} a_{\text{FIFO}(i_2)}^2(t) dt}{\int_{t_0^{\text{GAME}_{\text{follower}}(i_2)}}^{t_1^{\text{GAME}_{\text{follower}}(i_2)}} a_{\text{GAME}_{\text{follower}}(i_2)}^2(t) dt} + w_2 \frac{T_{\text{FIFO}(i_2)}}{T_{\text{GAME}_{\text{follower}}(i_2)}} \quad (24)$$

$$n_1 = w_1 \frac{\int_{t_0^{\text{FIFO}(i_1)}}^{t_1^{\text{FIFO}(i_1)}} a_{\text{FIFO}(i_1)}^2(t) dt}{\int_{t_0^{\text{GAME}_{\text{follower}}(i_1)}}^{t_1^{\text{GAME}_{\text{follower}}(i_1)}} a_{\text{GAME}_{\text{follower}}(i_1)}^2(t) dt} + w_2 \frac{T_{\text{FIFO}(i_1)}}{T_{\text{GAME}_{\text{follower}}(i_1)}} \quad (25)$$

$$n_2 = w_1 \frac{\int_{t_0^{\text{FIFO}(i_2)}}^{t_1^{\text{FIFO}(i_2)}} a_{\text{FIFO}(i_2)}^2(t) dt}{\int_{t_0^{\text{GAME}_{\text{leader}}(i_2)}}^{t_1^{\text{GAME}_{\text{leader}}(i_2)}} a_{\text{GAME}_{\text{leader}}(i_2)}^2(t) dt} + w_2 \frac{T_{\text{FIFO}(i_2)}}{T_{\text{GAME}_{\text{leader}}(i_2)}} \quad (26)$$

3.5. Coefficient parameters selection for the cost function

Optimal control for motion planning guarantees that vehicles will merge without colliding. Further, we need to select a proper cost function to minimize total fuel consumption and total travel time. Eq. (9) gives the mathematical expression, and it is a bi-objective optimization problem. The first item of Eq. (9) is the surrogate of fuel consumption and would have the effect for trajectory smoothing for motion planning (Rios-Torres and Malikopoulos, 2017), and the second item is related to travel time. Usually, a solution whereby both objectives are optimized would not exist and we should make trade-offs between these objectives, represented by the parameters w_1 and w_2 . In this paper, we try to find an unbiased Pareto solution for this bi-objective optimization problem. Pareto solutions are nondominated solutions, which means further optimization for an objective is always at the cost of deterioration of other objectives (Boyd and Vandenberghe, 2004). A group of Pareto solutions is called the Pareto solutions set, and the hyperplane consisting of the Pareto solutions is called the Pareto front. For example, in Fig. 3, we want to minimize the two objectives A and B, and each point on the line is a feasible solution. (x_1, y_1) is a Pareto solution, because if we continue to reduce the value near x_1 to optimize objective A, performance of objective B will deteriorate, and vice versa. (x_2, y_2) is also a Pareto solution and the analysis for this point is similar to that of (x_1, y_1) . (x_3, y_3) is not a Pareto solution, since both objectives can further decrease near x_3 . The blue section of the solid line is part of the Pareto front and any point on it can be a non-inferior solution for a bi-objective optimization problem.

From the above statement, we can conclude that the expected solution is on the Pareto front. For the cost function in

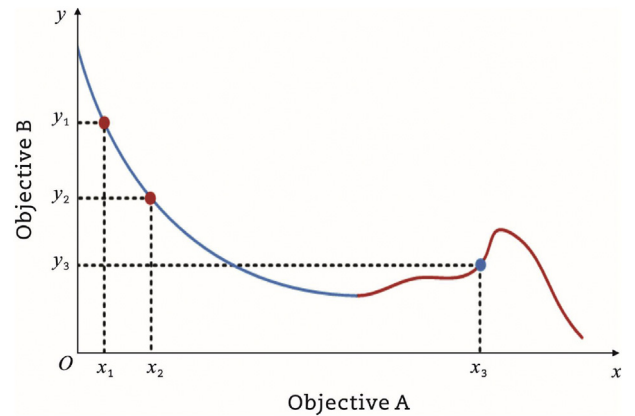


Fig. 3 – Explanation of the Pareto solutions.

Eq. (9), it consists of two items. The first item is related to the cumulative fuel consumption of all W vehicles during the research time, noted as F . The second item is the total travel time of all W vehicles, noted as T . Given different w_1 and w_2 , values of F and T are different, and our purpose is to determine the w_1 and w_2 corresponding to the least bias difference (Wang et al., 2017) between these two objectives.

For the estimation of fuel consumption F , we refer to Kamal et al. (2012), and it can be written as

$$\begin{cases} f_i(t) = f_{iv}(t) + f_{ia}(t) \\ f_{iv}(t) = q_0 + q_1 v_i(t) + q_2 v_i^2(t) + q_3 v_i^3(t) \\ f_{ia}(t) = a_i(t)(r_0 + r_1 v_i(t) + r_2 v_i^2(t)) \end{cases} \quad (27)$$

where $f_i(t)$ is the instantaneous fuel consumption of vehicle V_i , consisting of $f_{iv}(t)$ and $f_{ia}(t)$, and representing instantaneous fuel consumption caused by velocity and acceleration, respectively. If the acceleration value is negative, we use its absolute value instead. Coefficient vectors $q = [q_0, q_1, q_2]$ and $r = [r_0, r_1, r_2]$ are obtained from Zhao et al. (2018) and the values are $q = [1.569 \times 10^{-1}, 2.45 \times 10^{-2}, -7.415 \times 10^{-4}]$, and $r = [7.224 \times 10^{-2}, 9.681 \times 10^{-2}, 1.075 \times 10^{-3}]$. The total fuel consumption is the sum of instantaneous fuel consumption of all vehicles from t_i^0 to t_i^f , i.e.

$$F = \sum_i^W \int_{t_i^0}^{t_i^f} f_i(t) dt \quad (28)$$

To simplify the computation, we sample N points from t_i^0 to t_i^f and each sampling moment is noted as j , then Eq. (28) can be rewritten as

$$F = \sum_i^W \sum_j^N f_i(j) \quad (29)$$

where $f_i(j)$ is the instantaneous fuel consumption of vehicle V_i at sampling moment j . Total travel time T can be calculated as

$$T = \sum_i^W T_i = \sum_i^W (t_i^f - t_i^0) \quad (30)$$

Given different w_1 and w_2 , the cost function in Eq. (9) would be influenced so that the MS allocation to each vehicle is different and the values of F and T also change. Wang et al. (2017) proved that an unbiased Pareto solution that is acceptable for every objective can be determined for a multi-objective optimization problem if every objective is considered equally. They verified this conclusion in bi-objective optimization problems and pointed out that an ideal solution in the real sense would be the solution on the Pareto front, where the difference between the bias of each objective is the minimum.

In this paper, we design a numerical approach called varying-scale grid search to determine w_1 and w_2 in correspondence to the principle of least bias difference between F and T , described in Algorithm 1. For step 3 in each loop (representing different searching step length, i.e., different scale), we adopt the method designed by Tom (2020), which uses modified quicksort to find the Pareto front. This method has

a high efficiency in dealing with the problem of determining the Pareto front of the discrete points. For step 4 in each loop, we adopt the algorithm proposed by Wang et al. (2017) (Appendix A) to determine the Pareto solution with least difference bias. This algorithm is based on strict and objective mathematic derivation so that the solution we finally obtained is theoretically optimal if we regard the two objectives in the bi-objective optimization problem as equally important.

Algorithm 1: varying-scale grid search for coefficient parameters w_1 and w_2 .

Input: initial searching step length σ_1, σ_2 ; scaling factor $\alpha_1, \alpha_2 \in (0, 1)$; grid precision ϵ .

Output: corresponding w_1, w_2 for the acceptable Pareto solution.

Steps:

1. set the initial searching range l_1, l_2 for w_1, w_2 , i.e., set the upper and lower bound for w_1, w_2 .

Loop

If $\sigma_1 > \epsilon$ && $\sigma_2 > \epsilon$:

2. divide l_1, l_2 with step length σ_1, σ_2 , respectively, and calculate F and T for each divided grid point (w_1, w_2) .

3. find the Pareto front using modified quicksort.

4. determine the point (F, T) on the Pareto front where the difference between the bias of each objective is the minimum.

5. search the corresponding $(w_1, w_2) \leftarrow (F, T)$.

6. $l_1 := [w_1 - \sigma_1, w_1 + \sigma_1]$, $l_2 := [w_2 - \sigma_2, w_2 + \sigma_2]$, $\sigma_1 := \alpha_1 \sigma_1$, $\sigma_2 := \alpha_2 \sigma_2$.

Else: end loop

End

4. Simulation results

To validate the availability and effectiveness of the proposed model, we have simulated the on-ramp merging scenario presented in the previous section. In this section, we first explain how we search the coefficient parameters in the cost function in Eq. (9), and then we display the profiles of merging CAVs along with a comparison with other models in terms of total fuel consumption and total travel time. With an increasing number of CAVs entering the control area, we set the vehicle number as 10 in our simulation as a case study to facilitate the computation of total fuel consumption and total travel time. For other simulation parameters, we referred to Chen et al. (2020) and set the length of the control area as $L = 400$ m (considering the communication range) and the length of the merging area as $M = 30$ m (length of the potential collision area). The speed of vehicles at the control line is randomly initialized and obey normal distribution. Specifically, the initial speed $v_i(t_i^0) \sim N(15, 0.5^2)$ (the unit for the speed is m/s). And all vehicles are expected to pass the merging area with constant speed equal to $v_m = 13.4$ m/s due to control effect of the centralized controller (Jing et al., 2019; Rios-Torres and Malikopoulos,

2017). Constraints of acceleration and velocity are set as $[-3, 3]$ m/s^2 and $[0, 30]$ m/s , respectively.

4.1. Determination of coefficient parameters for the cost function

According to Algorithm 1, we set the initial searching step length as $\sigma_1 = \sigma_2 = 0.05$ and the scaling factor as $\alpha_1 = \alpha_2 = 0.2$. The grid precision ϵ was set as 0.01 and the initial searching range l_1, l_2 for w_1, w_2 was set as $[0, 1]$, i.e. $l_1, l_2 \in [0, 1]$.

Under the condition that the two objectives in Eq. (9) (i.e., the surrogate total fuel consumption and total travel time) are equally important, we first divided the searching range $l_1, l_2 \in [0, 1]$ with searching step length $\sigma_1 = \sigma_2 = 0.05$ and calculated the surrogate total fuel consumption and total travel time at each grid point. Then candidates on the Pareto front were found via modified quicksort algorithm, marked with green circle in Fig. 4, and the point with least bias difference was selected as the best solution, marked with red star in Fig. 4, under the given search step. The corresponding w_1 and w_2 were 0.30 and 0.80, respectively. Then we narrowed the search range and set $l_1 \in [0.25, 0.35]$, $l_2 \in [0.75, 0.85]$ with more precise searching step length $\sigma_1 = \sigma_2 = 0.01$ and repeated the procedures outlined above to determine the optimal solution under the given search step. The result is shown in Fig. 5. Since the grid precision ϵ was satisfied, the searching process ended and the point marked in red star in Fig. 5 was the optimal solution. Its corresponding $w_1 = 0.32$ and $w_2 = 0.79$ were the final coefficient parameters we selected.

4.2. Profiles of CAVs merging

Once an MS is determined by complete information static game, acceleration, velocity, and position profiles can be obtained through optimal control. Figs. 6–8 show vehicle profiles during research time, where we set the number of vehicles $W = 10$ and constrain acceleration to $[-3, 3]$ m/s^2 and velocity to $[0, 30]$ m/s . Expected time headway when vehicles pass the merging area is $M/v_m = 2.26$ s.

From the simulation results, we can see that all the constraints are satisfied, and all vehicles can cooperatively merge at the ramp. Results shown in Fig. 7 illustrate that even though the initial velocity of each vehicle is different, they can adjust their speeds to the expected merging speed v_m via the optimal control. Fig. 8 validates that the centralized controller coordinates each vehicle passing the merging area only after the previous vehicle has left. All vehicles keep a constant distance equal to the length of merging area $M = 30$ m and keep the time headway which equals the expected time headway M/v_m .

4.3. Comparison of total fuel consumption and total travel time with other models

To validate the effectiveness of the proposed model on total fuel consumption and total travel time, we compared our model with the FIFO model and the stop model. For the FIFO model, as mentioned above, an MS is allocated only at the moment when a vehicle reaches the control line; i.e., the earlier the vehicle reaches the control line, the higher the priority that its MS order will be allocated. The motion planning method (i.e., optimal control) is the same for the proposed method and the FIFO model in this context. For the stop model, we adopt the baseline scenario depicted by Rios-Torres and Malikopoulos (2017), and the car-following model for CAVs proposed by Li and Ma (2017) to describe the vehicles' behaviors on the ramp road. In such scenario, vehicles on the ramp road have to stop before the intersection to allow the vehicles on the main road to cross the merging area. Only after all the vehicles on the main road have passed the ramp, the leading vehicle on the ramp road start accelerating to reach again the maximum allowed speed and the behaviors of other vehicles on the ramp road are constrained by the car following model.

Total fuel consumption and total travel time can be calculated with Eqs. (29) and (30), respectively. Velocity and acceleration constraints and related settings (for instance, vehicle number $W = 10$, length of the control area $L = 400$ m, length of the merging area $M = 30$ m, initial speed at the

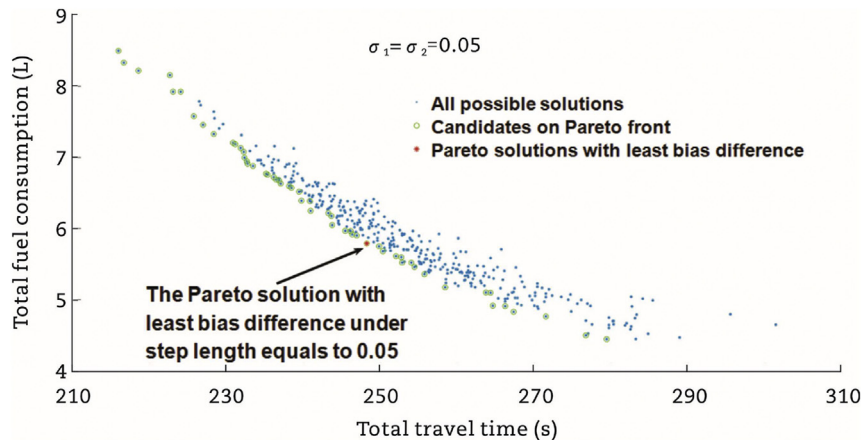


Fig. 4 – Grid search under step length equal to 0.05.

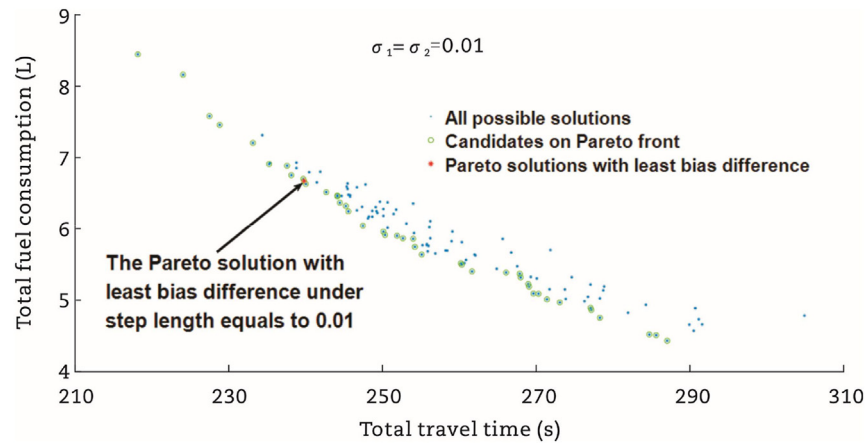


Fig. 5 – Grid search under step length equal to 0.01.

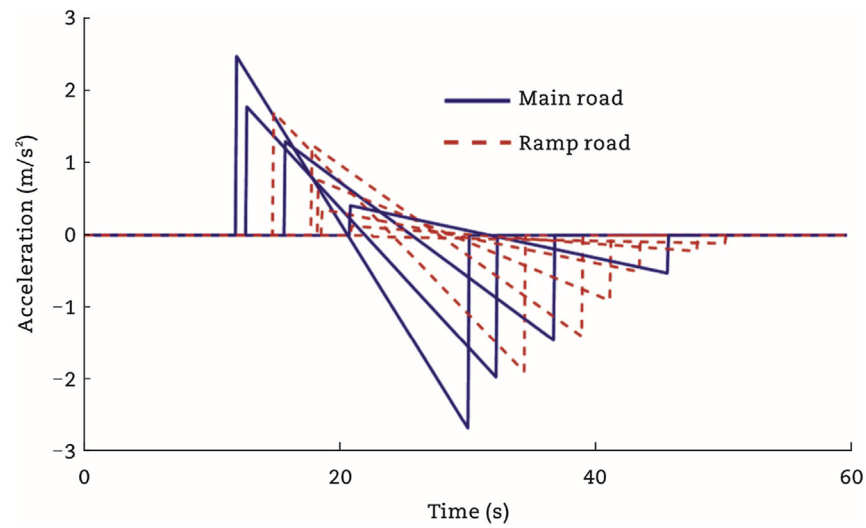


Fig. 6 – Control inputs (acceleration) of the vehicles.

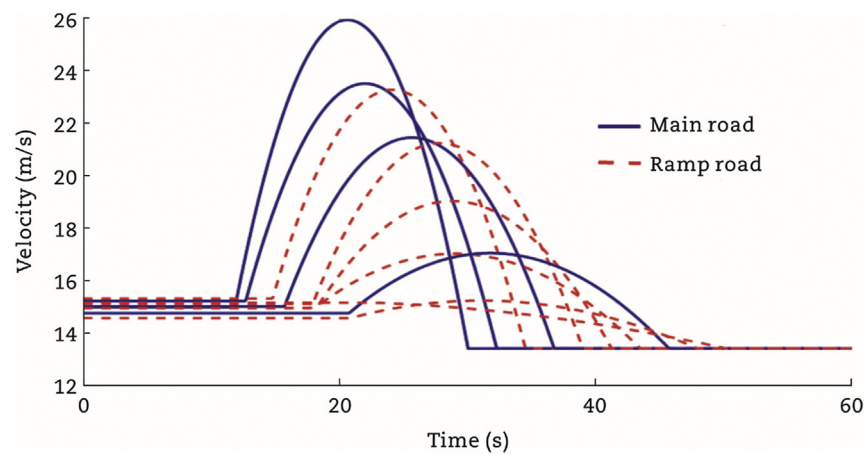


Fig. 7 – Velocity profiles of the vehicles.

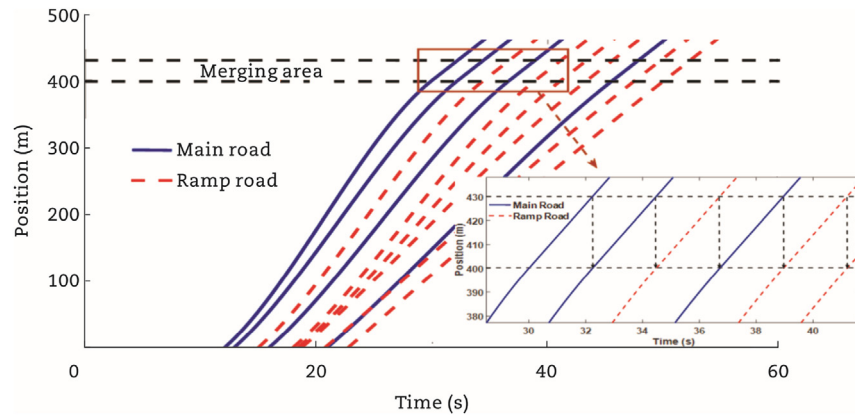


Fig. 8 – Trajectory profiles of the vehicles (control line as the start).

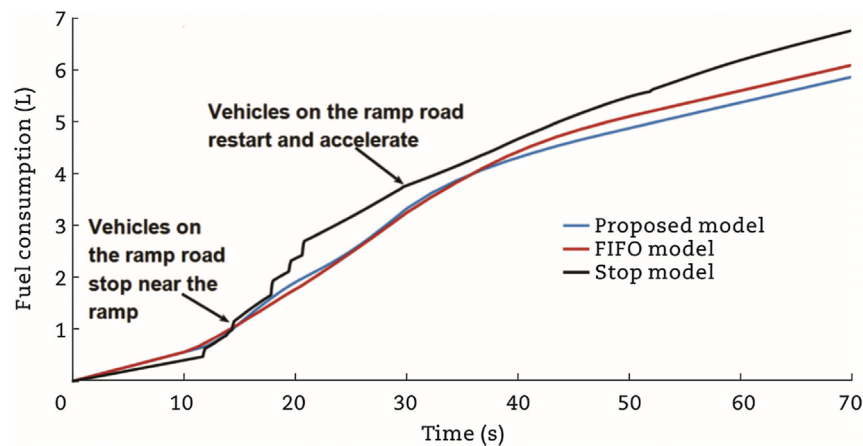


Fig. 9 – Comparison of total fuel consumption between different models.

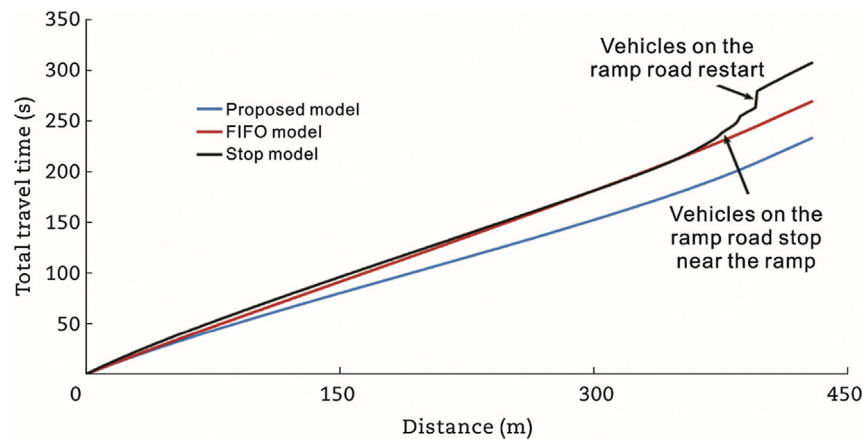


Fig. 10 – Comparison of total travel time between different models.

Table 2 – Performance improvement of the proposed model.

Compared model	Percentage of the total fuel consumption reduction (%)	Percentage of the total travel time saved (%)
FIFO model	3.63	11.54
Stop model	13.20	17.00

control line and merging velocity $v_m = 13.4$ m/s) are the same in these three models. Results are shown in Figs. 9 and 10.

From Figs. 9 and 10, we can see that cumulative fuel consumption and total travel time are similar in the proposed model and the FIFO model, since many vehicles have the same MS in both models (Fig. 8). While, the proposed model performs better in terms of total fuel consumption and total

travel time at the end. Significantly, an MS obtained from the proposed model for each vehicle can be the same as that from the FIFO model. In this situation, these two models are equivalent. When the MS obtained from these two models is different, the proposed model is more efficient, since it takes more into account about the state of the vehicles. The stop model is undoubtedly the worst of these three models, in terms of total fuel consumption and total travel time. In Fig. 9, we can see that stopping and restarting vehicles on the ramp road leads to increasing cumulative fuel consumption. Fig. 10 shows that vehicles on the ramp road stopping near the ramp leads to a significant increase in total travel time. Compared with the stop model and FIFO model, the proposed model reduces the total fuel consumption and saves the total travel time in this simulation where we set the vehicle number as 10. Detailed results are shown in Table 2.

5. Conclusions

In this paper, we study the scenario of on-ramp CAVs merging, whereby each vehicle is modeled as a mass point and can communicate with a centralized controller without any time delay. The core problems in this scenario are the allocation of MS and motion planning for each vehicle. To solve these, we first propose an MS allocation method based on a complete information static game. Then we adopt optimal control based on Pontryagin's minimum principle to solve the motion planning problem. Further, we discuss the approach to determining key parameters, including the pay-off of each player in the pay-off matrix and the coefficient parameters for the cost function. Calculation of the pay-off for each player in different situations uses the FIFO model as a benchmark, and coefficient parameters are obtained via the proposed varying-scale grid search method, where grid search with different step length is used to search for possible solutions at various scales. A modified quicksort algorithm is designed to find the Pareto front, and finally, the (approximately) unbiased Pareto solution is determined as the optimal solution. We simulate the scenario and illustrate the process of determining the optimal solution on the Pareto front under the situation that both objectives are equally important for a bi-objective optimization problem. The profiles of merging CAVs and comparison with other models in terms of total fuel consumption and total travel time are also presented to validate the availability and effectiveness of the proposed model.

In this paper, we constrain that only one vehicle can be in the merging area at a time. This constraint is a sufficient condition to avoid potential lateral collision. However, considering the scenario that a consecutive string of vehicles on one of the roads are traveling through the merging point (there is no potential lateral collision), such constraint can be casted. In such case, Eq. (13) is invalid and the final value condition t_i^f for each vehicle should be reconsidered to solve Eq. (21), which increase the difficulty to obtain the solution. In addition, there are several ideal assumptions in this paper and one of these assumptions is that we model all vehicles

as mass point. If the length and width of the vehicles are considered, lateral control methods should be added to avoid collision. Besides, communication delay is ignored in our model and such delay may influence the MS allocation and motion planning. Therefore, the lateral control of the vehicles on the ramp and the influence of the time delay would be further studied in the future work.

Conflicts of interest

The authors do not have any conflict of interest with other entities or researchers.

Acknowledgments

This work is supported in by National Natural Science Foundation of China (No.61903046), Key Research and Development Program of Shaanxi Province (No.2021GY-290), Youth Talent Lift Project of Shaanxi Association for Science and Technology (No.20200106), and Joint Laboratory for Internet of Vehicles, Ministry of Education-China Mobile Communications Corporation (No.213024170015), Fundamental Research Funds for the Central Universities (No. 300102240106).

Appendix A. Explanation to the step 4 of Algorithm 1

Here, we briefly illustrate how to determine the Pareto solution with least bias difference for a bi-objective optimization problem. Readers can obtain more details from Wang et al. (2017).

It can be observed that the distribution of the Pareto front monotonically increases or decreases. Assuming that we have M solutions on the Pareto front, $f_1^{(m)}$ ($m = 1, 2, \dots, M$) is the function value for objective one (for example, in this paper, the total travel time T) in ascending order, and $f_2^{(m)}$ ($m = 1, 2, \dots, M$) is the function value for objective two (for example, in this paper, the total fuel consumption F) corresponding to the value of $f_1^{(m)}$. Because of the monotonicity and the features of the Pareto front, it must be in descending order.

Step 1: compute the average variability $k_1^{(m)}$, $k_2^{(m)}$.

$$k_1^{(m)} = \frac{1}{2} \left(\frac{f_2^{(m)} - f_2^{(m-1)}}{f_1^{(m)} - f_1^{(m-1)}} + \frac{f_2^{(m+1)} - f_2^{(m)}}{f_1^{(m+1)} - f_1^{(m)}} \right) \quad m = 2, 3, \dots, M-1 \quad (A1)$$

$$k_2^{(m)} = \frac{1}{2} \left(\frac{f_1^{(m)} - f_1^{(m-1)}}{f_2^{(m)} - f_2^{(m-1)}} + \frac{f_1^{(m+1)} - f_1^{(m)}}{f_2^{(m+1)} - f_2^{(m)}} \right) \quad m = 2, 3, \dots, M-1 \quad (A2)$$

Step 2: compute the sensitivity ratio of average variability to the corresponding value of the objective function δ_1^m, δ_2^m .

$$\delta_1^m = \frac{k_1^{(m)}}{f_1^{(m)}} \quad m = 2, 3, \dots, M-1, f_1^{(m)} \neq 0 \quad (A3)$$

$$\delta_2^m = \frac{k_2^{(m)}}{f_2^{(m)}} \quad m = 2, 3, \dots, M-1, f_2^{(m)} \neq 0 \quad (A4)$$

Step 3: determine pareto solution with least bias difference for a bi-objective optimization problem.

Define

$$\epsilon_1^m = \frac{\delta_1^m}{\sum_{i=2}^{M-1} \delta_1^{(i)}} \quad m = 2, 3, \dots, M-1 \quad (A5)$$

$$\epsilon_2^m = \frac{\delta_2^m}{\sum_{i=2}^{M-1} \delta_2^{(i)}} \quad m = 2, 3, \dots, M-1 \quad (A6)$$

$$\Delta\epsilon^{(m)} = |\epsilon_1^{(m)} - \epsilon_2^{(m)}| \quad m = 2, 3, \dots, M-1 \quad (A7)$$

$$(\Delta\epsilon)_{\min} = \min\{\Delta\epsilon^{(2)}, \Delta\epsilon^{(3)}, \dots, \Delta\epsilon^{(M-1)}\} \quad (A8)$$

The solution corresponding to $(\Delta\epsilon)_{\min}$ is the (approximately) unbiased solution, or in another word, the optimal Pareto solution with least bias difference.

REFERENCES

- Altan, O.D., Wu, G., Barth, M.J., et al., 2017. GlidePath: eco-friendly automated approach and departure at signalized intersections. *IEEE Transactions on Intelligent Vehicles* 2 (4), 266–277.
- Awal, T., Kulik, L., Ramamohanrao, K., 2013. Optimal traffic merging strategy for communication and sensor-enabled vehicles. In: 16th International IEEE Conference on Intelligent Transportation Systems, The Hague, 2013.
- Boyd, S.P., Vandenberghe, L., 2004. *Convex Optimization*. Cambridge University Press, Cambridge.
- Cao, W., Mukai, M., Kawabe, T., et al., 2015. Cooperative vehicle path generation during merging using model predictive control with real-time optimization. *Control Engineering Practice* 34, 98–105.
- Chen, L., Englund, C., 2015. Cooperative intersection management: a survey. *IEEE Transactions on Intelligent Transportation Systems* 17 (2), 570–586.
- Chen, N., van Arem, B., Alkim, T., et al., 2020. A hierarchical model-based optimization control approach for cooperative merging by connected automated vehicles. *IEEE Transactions on Intelligent Transportation Systems*, <https://doi.org/10.1109/TITS.2020.3007647>.
- Chung, Y., Recker, W.W., 2013. Spatiotemporal analysis of traffic congestion caused by rubbernecking at freeway accidents. *IEEE Transactions on Intelligent Transportation Systems*, 14 (3), 1416–1422.
- Hu, Y., Nakhaei, A., Tomizuka, M., et al., 2019. Interaction-aware decision making with adaptive strategies under merging scenarios. In: 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Macau, 2019.
- Iordanidou, G.R., Roncoli, C., Papamichail, I., et al., 2014. Feedback-based mainstream traffic flow control for multiple bottlenecks on motorways. *IEEE Transactions on Intelligent Transportation Systems* 16 (2), 610–621.
- Jing, S., Hui, F., Zhao, X., et al., 2019. Cooperative game approach to optimal merging sequence and on-ramp merging control of connected and automated vehicles. *IEEE Transactions on Intelligent Transportation Systems* 20 (11), 4234–4244.
- Kabashkin, I., 2017. Reliable V2X communications for safety-critical intelligent transport systems. In: 2017 Advances in Wireless and Optical Communications (RTUWO), Riga, 2017.
- Kang, K., Rakha, H.A., 2017. Game theoretical approach to model decision making for merging maneuvers at freeway on-ramps. *Transportation Research Record* 2623, 19–28.
- Kamal, M.A.S., Mukai, M., Murata, J., et al., 2012. Model predictive control of vehicles on urban roads for improved fuel economy. *IEEE Transactions on Control Systems Technology* 21 (3), 831–841.
- Li, L., Ma, W., 2017. A collision-free car-following model for connected automated vehicles. In: 96th Transportation Research Board Annual Meeting, Washington DC, 2017.
- Li, S., Zheng, Y., Li, K., et al., 2017. Dynamical modeling and distributed control of connected and automated vehicles: challenges and opportunities. *IEEE Intelligent Transportation Systems Magazine* 9 (3), 46–58.
- Lin, Y., McPhee, J., Azad, N.L., 2020. Anti-jerk on-ramp merging using deep reinforcement learning. In: 2020 IEEE Intelligent Vehicles Symposium, Las Vegas, 2020.
- Matarage, I.C., Dissanayake, S., 2020. Quality assessment between calibrated highway safety manual safety performance functions and calibration functions for predicting crashes on freeway facilities. *Journal of Traffic and Transportation Engineering (English Edition)* 7 (1), 76–87.
- Massera Filho, C., Terra, M.H., Wolf, D.F., 2017. Safe optimization of highway traffic with robust model predictive control-based cooperative adaptive cruise control. *IEEE Transactions on Intelligent Transportation Systems* 18 (11), 3193–3203.
- Min, H., Wu, X., Cheng, C., et al., 2019. Kinematic and dynamic vehicle model-assisted global positioning method for autonomous vehicles with low-cost GPS/camera/in-vehicle sensors. *Sensors* 19 (24), 5430.
- Moradi-Pari, E., Mahjoub, H.N., Kazemi, H., et al., 2017. Utilizing model-based communication and control for cooperative automated vehicle applications. *IEEE Transactions on Intelligent Vehicles* 2 (1), 38–51.
- Morales, A., Nijmeijer, H., 2016. Merging strategy for vehicles by applying cooperative tracking control. *IEEE Transactions on Intelligent Transportation Systems* 17 (12), 3423–3433.
- Myerson, R.B., 1997. *Game Theory: Analysis of Conflict*. Harvard University Press, Cambridge.
- Ntousakis, I.A., Porfyri, K., Nikolos, I.K., et al., 2014. Assessing the impact of a cooperative merging system on highway traffic using a microscopic flow simulator. In: ASME 2014 International Mechanical Engineering Congress and Exposition, Montreal, 2014.
- Ntousakis, I.A., Nikolos, I.K., Papageorgiou, M., 2016. Optimal vehicle trajectory planning in the context of cooperative merging on highways. *Transportation Research Part C: Emerging Technologies* 71, 464–488.
- Orbay, B.Z., Sevgi, L., 2014. Game theory and engineering applications. *IEEE Antennas and Propagation Magazine* 56 (3), 256–267.
- Raravi, G., Shingde, V., Ramamritham, K., et al., 2007. Merge algorithms for intelligent vehicles. In: Ramesh, S., Sampath, P. (Eds.), *Next Generation Design and Verification Methodologies for Distributed Embedded Control Systems*. Springer, Dordrecht, pp. 51–65.
- Rios-Torres, J., Malikopoulos, A.A., 2016. A survey on the coordination of connected and automated vehicles at intersections and merging at highway on-ramps. *IEEE Transactions on Intelligent Transportation Systems* 18 (5), 1066–1077.
- Rios-Torres, J., Malikopoulos, A.A., 2017. Automated and cooperative vehicle merging at highway on-ramps. *IEEE Transactions on Intelligent Transportation Systems* 18 (4), 780–789.
- Tom, R., 2020. Find Multi-Objective Pareto Front Using Modified Quicksort. Available at: <https://www.mathworks.com/>

matlabcentral/fileexchange/73089-find-multi-objective-pareto-front-using-modified-quicksort (Accessed 18 April 2021).

- Triest, S., Villaflor, A., Dolan, J.M., 2020. Learning highway ramp merging via reinforcement learning with temporally-extended actions. In: 2020 IEEE Intelligent Vehicles Symposium (IV), Las Vegas, 2020.
- Wang, M., 2018. Infrastructure assisted adaptive driving to stabilise heterogeneous vehicle strings. *Transportation Research Part C: Emerging Technologies* 91, 276–295.
- Wang, N., Zhao, W.J., Wu, N., et al., 2017. Multi-objective optimization: a method for selecting the optimal solution from Pareto non-inferior solutions. *Expert Systems with Applications* 74, 96–104.
- Wang, Y., E, W., Tang, W., et al., 2013. Automated on-ramp merging control algorithm based on internet-connected vehicles. *IET Intelligent Transport Systems* 7 (4), 371–379.
- Wang, Z., Wu, G., Barth, M., 2018. Distributed consensus-based cooperative highway on-ramp merging using V2X communications. SAE Technical Paper, <https://doi.org/10.4271/2018-01-1177>.
- Yang, M., Ai, B., He, R., et al., 2018. Influence of different antenna locations on channel characterization for V2V communications. In: 2018 IEEE International Symposium on Antennas and Propagation & USNC/URSI National Radio Science Meeting, Boston, 2018.
- Zhao, X., Min, H., Xu, Z., et al., 2019. An ISVD and SFFSD-based vehicle ego-positioning method and its application on indoor parking guidance. *Transportation Research Part C: Emerging Technologies* 108, 29–48.
- Zhao, X., Sun, P., Xu, Z., et al., 2020. Fusion of 3D LIDAR and camera data for object detection in autonomous vehicle applications. *IEEE Sensors Journal* 20 (9), 4901–4913.
- Zhao, W., Ngoduy, D., Shepherd, S., et al., 2018. A platoon based cooperative eco-driving model for mixed automated and human-driven vehicles at a signalised intersection. *Transportation Research Part C: Emerging Technologies* 95, 802–821.



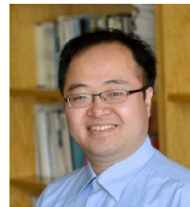
Haigen Min (member, IEEE) received the B.S., M.S., and PhD degrees from the Department of Traffic Information Engineering and Control, Chang'an University, China. He is currently a lecturer in School of Information Engineering at Chang'an University. His research interests include high-precision localization, environment perception, and cooperative control for connected and automated vehicles and its fault detection and diagnosis.



Yukun Fang received the B.E. degree in communication engineering from Chang'an University, China, in 2019, where he is currently pursuing the PhD degree in traffic information engineering and control in School of Information Engineering. His current research interests include control theories for autonomous vehicles, fault detection and diagnosis and game theory.



Xia Wu received the B.E. degree in computer science from Chang'an University, China, in 2015, where she is currently pursuing the PhD degree in traffic information engineering and control in School of Information Engineering. Her current research interests include mixed traffic flow control, trajectory optimization in a mixed traffic flow with CAVs and HDVs.



Guoyuan Wu received his PhD degree in mechanical engineering from the University of California, Berkeley in 2010. Currently, he holds an associate researcher and an associate adjunct professor position in Bourns College of Engineering-Center for Environmental Research and Technology (CE-CERT) and Department of Electrical and Computer Engineering at University of California at Riverside. His research focuses on development and evaluation of sustainable and intelligent transportation system technologies, including connected and automated transportation systems, shared mobility, transportation electrification, optimization and control of vehicles, traffic simulation, and emissions measurement and modeling. Dr. Wu is an associate editor of SAE Journal-Connected and Automated Vehicles and IEEE Open Journal of Intelligent Transportation Systems. He is also a member of the Vehicle-Highway Automation Standing Committee of the Transportation Research Board (TRB), a board member of Chinese Institute of Engineers Southern California Chapter (CIE-SOCAL), and a member of Chinese Overseas Transportation Association (COTA). He is a recipient of Vincent Bendix Automotive Electronics Engineering Award.



Xiangmo Zhao (member, IEEE) received the PhD degree from Chang'an University, Xi'an, China. He is currently a professor in School of Information Engineering, Chang'an University. He is a vice president of the Joint Laboratory for Connected vehicles, Ministry of Education and China Mobile, and the Shaanxi Road Traffic Intelligent Detection and Equipment Engineering Technology Research Centre, and is also a leader of the National Key Subjects-Traffic Information Engineering and Control, Chang'an University. His current research interests include connected vehicles, automated vehicles, intelligent transportation systems, and computer science. He is the director of the Information Professional Committee, a member of the Advisory Expert Group of the China Transportation Association, a member of the National Motor Vehicle Operation Safety Testing Equipment Standardization Committee and the Leading Group of the National Traffic Computer Application Network, the vice chairman of the Institute of Highway Association on Computer Professional Committee, and the deputy director of the Institute of Computer in Shaanxi Province.